

Akhil Joshi

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SUMMARY

Computer Science undergraduate specializing in AI/ML, backend architecture, and agentic workflows. Proven hands-on experience in building scalable FastAPI microservices, LLM tool integrations, and automated data pipelines. Passionate about rapid prototyping, conversational AI agents, and collaborating directly with stakeholders to translate real-world requirements into functional AI deployments.

EDUCATION

Vellore Institute of Technology

B.Tech, Computer Science Engineering (AI & Machine Learning)

Bhopal, India

2023 – Present

TECHNICAL SKILLS

Languages: Python, JavaScript, Java, SQL

AI & Agentic Skills: Prompt Engineering, LLM APIs, FastAPI, Function Calling / Tool Use, Model Context Protocol (MCP)

Backend & APIs: RESTful APIs, Node.js, Express.js, Flask, OAuth2.0/JWT Security, API Management

Cloud & DevOps: Docker, CI/CD Pipelines, Render, Azure (Storage, App Services, Functions)

Databases: MySQL, MongoDB, Relational Schema Design

EXPERIENCE

IT Intern

Taj Cement (Hills Cement Company Ltd.)

May 2026 – June 2026

Bhopal, India

- Engineered optimized SQL backend queries and data processing pipelines across 3 internal business units, reducing database retrieval latency by 25%.
- Developed backend automation scripts that processed 100,000+ operational records, cutting manual execution time by 20% and improving system throughput by 15%.
- Collaborated with 4 cross-functional technical teams to analyze operational workflows, document requirements, and support secure data integration across systems.

PROJECTS

AI MCP Tool Server | *Python, FastAPI, FastMCP, Docker, CI/CD*

- Developed a scalable Model Context Protocol (MCP) server enabling systems to securely connect, execute function-calling/tool-use commands, and consume external services via robust API endpoints.
- Implemented comprehensive API management through JWT authentication and rate limiting, deploying via automated CI/CD pipelines on Render.

Insight AI - Enterprise Analytics Platform | *FastAPI, PostgreSQL, React, OAuth2.0, Docker*

- Engineered a secure full-stack data platform integrating LLM-driven analytics workflows, implementing zero-trust database security and JWT/OAuth2.0 authentication.
- Built and exposed clean RESTful APIs connecting frontend inputs directly to modular backend services for real-time customer data processing.

SentinelIQ IoT Microservices Platform | *Python, Flask, Docker, REST API, Machine Learning*

- Architected a cloud-native microservices backend containerized with Docker to simulate, consume, and process real-time telemetry streams from 20+ industrial sensors.
- Developed lightweight REST API routes for low-latency operational monitoring, optimizing data pipelines for resilient cloud interactions.

CERTIFICATIONS & ACHIEVEMENTS

Microsoft Azure Certification: AZ-900 Microsoft Azure Fundamentals

HackerRank Certifications: 5-Star Java Badge (Top global performer in OOP); SQL Intermediate Certified

Hackathons: Adobe Hackathon (Top 45 Teams among 1 Lakh+ teams); Devfolio Hackathon (Top 50 Teams among 500+ teams)